A Mathematical Theory of Economic Agents: Receivers, Floors, and the Algebra of Bounded Inquiry

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Abstract

We develop a mathematical theory of economic agents from a single primitive: the *receiver*, a four-tuple (K, D, Π, β) consisting of a knowledge framework, a decoder, a candidate-projection, and a strictly positive noise floor. From this primitive we derive, as theorems rather than assumptions, the following: (i) the Floor Theorem — the irreducible lower bound $\beta > 0$ on any receiver's semantic distance to a target; (ii) the Cell-Truth Theorem — all states within the same action-cell are S-indistinguishable; (iii) Mode Non-Privilege — the priority of fast pre-decoder layers is a theorem of optimality, not a heuristic assumption; (iv) a Banach fixed-point characterisation of every methodology's floor; (v) the Agent Triple (R, M, G) and its composite floor; (vi) bounded rationality as a mathematical theorem; (vii) eleven classical prerequisites for point-meaning each individually entail $\beta = 0$, which the Floor Theorem forbids; (viii) the *Gödelian residue* — private information irreducible below β — as a structural property of any bounded receiver; and (ix) Arrow's impossibility, the Grossman–Stiglitz paradox, and the exploration–exploitation incompatibility as corollaries. All results are proved in full. Thirty-five numerical experiments verify every quantitative prediction to machine precision ($\leq 10^{-14}$).

Contents

1 Introduction

1.1 The agent problem in economics

The canonical economic agent is specified by a utility function $U : X \to \mathbb{R}$ defined over a set of outcomes X, together with a complete and transitive preference ordering $\succeq$ over X induced by the relation $x \succeq x' \iff U(x) \geq U(x')$. Behaviour is then the maximisation of U subject to a budget or technological constraint. This architecture, developed axiomatically by von Neumann and Morgenstern [32] and extended by Savage [26] to subjective probability, underlies virtually all of positive economics.

Three foundational difficulties attend this specification.

Preferences are assumed, not derived. The ordering $\succeq$ is taken as a primitive datum of the model. No mechanism is given for *generating* preferences from an underlying representation system. When preferences conflict across contexts or agents, the standard apparatus provides no principled means of resolution.

Bounded rationality is a behavioral patch. Simon [28, 30] documented that real agents do not globally maximise but rather *satisfice*: they search sequentially through alternatives and halt at the first option that meets an aspiration level. The standard response has been to append "bounds" to the rational-choice model — computational limits, search costs, attention constraints — as auxiliary hypotheses. These bounds are behaviorally motivated; they are not derived from the agent's representational structure.

Substrate neutrality is lacking. A human consumer, a firm, and an algorithmic trading system receive the same formal apparatus: a utility function and a constraint set. Nothing in the standard theory explains why entities with radically different computational substrates and representational structures should share an identical mathematical description.

Point-valued utility requires infinite precision. The function $U : X \to \mathbb{R}$ assigns a unique real number to every outcome state. This presupposes that every state is uniquely identifiable — a form of infinite representational precision that we show is impossible for any bounded agent.

The present paper addresses all four difficulties by deriving the agent from a simpler primitive: the *receiver*. We show that bounded rationality, selective rationality, the impossibility of point-values, the irreducibility of private information, and the structural limits of value theory are all theorems that follow from the receiver's definition. They are not modifications of the standard model; they replace it.

1.2 Main contributions

The contributions of this paper are as follows.

(i) **Receiver primitive and Floor Theorem** (Sections 3 and 4). We introduce the receiver (K, D, Π, β) and prove that $S_\flat(R) = \beta > 0$ is the irreducible lower bound on the S-functional for any bounded receiver.

(ii) **Cell-Truth Theorem** (Section 5). All states within the same action-cell attain $S = \beta$ and are therefore S-indistinguishable. Representational invariance across isometric encodings follows as a corollary.

(iii) **Mode Non-Privilege and Selective Rationality** (Section 6). For a layered receiver, the pre-decoder layer (System 1) has strict priority whenever it suffices. This is a theorem of optimality, not a behavioral assumption.

(iv) **Methodology Banach Fixed Point** (Section 7). Every methodology $M = (T, \kappa, \sigma)$ has a unique attractor $S_\flat(M) = \sigma\kappa/(1-\kappa) > 0$ that is the irreducible methodological floor.

(v) **Agent Triple** (Section 8). The agent $A = (R, M, G)$ has floor $S_\flat(A) = \beta \cdot \sigma\kappa/((1-\kappa)\Sigma)$. The construction–action exclusion principle follows.

(vi) **Bounded Rationality as Theorem** (Section 4). No bounded receiver can achieve $S = 0$. Simon's satisficing is optimal by Theorem 6.4.

(vii) **Eleven-fold collapse to $\beta = 0$** (Section 9). Eleven classical prerequisites for point-meaning each individually require $\beta = 0$, which the Floor Theorem forbids.

(viii) **Gödelian residue = private information = β** (Section 10). The residual content of any solution that the agent cannot communicate is exactly the floor β.

(ix) **Thirty-five numerical experiments** (Section 12). All quantitative predictions are verified to relative error $\leq 10^{-14}$.

1.3 Organisation

Section 2 introduces outcome spaces and action-cells. Section 3 defines the receiver and the S-functional. Section 4 proves the Floor Theorem and establishes bounded rationality. Section 5 proves Cell-Truth and Representational Invariance. Section 6 treats layered receivers and selective rationality. Section 7 develops the methodology Banach fixed point. Section 8 defines the agent triple and the uncertainty principle. Section 9 proves the eleven-fold collapse and the cell-value theorems. Section 10 treats the Gödelian residue and private information. Section 11 reconsiders classical impossibility results. Section 12 presents the numerical validation. Section 13 situates the theory in the existing literature. Section 14 concludes.

2 Outcome Spaces and Action-Cells

Definition 2.1 (Outcome space)**.** An *outcome space* is a triple (X, d, Σ) where

(a) X is a nonempty set of *outcome states*;

(b) $d\colon X \times X \to [0, \Sigma]$ is a *bounded pseudometric*: $d(x, x) = 0$ for all x; $d(x, x') = d(x', x)$; and $d(x, x'') \leq d(x, x') + d(x', x'')$ for all $x, x', x'' \in X$;

(c) $\Sigma \in (0, \infty)$ is the *maximal semantic distance.*

The canonical normalisation is $\Sigma = 100$; no result in this paper depends on the specific value.

Definition 2.2 (Action map)**.** An *action map* is a measurable surjection $\mathrm{Act}\colon X \to Y$, where (Y, d_Y) is a metric space called the *action space.* For any outcome state $x \in X$, the element $\mathrm{Act}(x) \in Y$ is the *practical-response class* of x.

Definition 2.3 (Action-cell)**.** For $y \in Y$ the *action-cell* of y is the preimage

$$C_y \;:=\; \mathrm{Act}^{-1}(\{y\}) \;\subseteq\; X.$$

Each action-cell is equipped with three numerical invariants:

(a) *Tolerance*: $\tau(C_y) := \sup_{x\in X} \inf_{c\in C_y} d(x, c) > 0$.

(b) *Cell distance*: $d(x, C_y) := \inf_{c\in C_y} d(x, c)$.

(c) *Diameter*: $\mathrm{diam}(C_y) := \sup_{c,c'\in C_y} d(c, c') < \Sigma$.

A *generic action-cell* is any nonempty $C \subseteq X$ with $\tau(C) > 0$ and $\mathrm{diam}(C) < \Sigma$.

Proposition 2.4 (Metric properties of cell distance)**.** *For any generic action-cell $C \subseteq X$, the function $x \mapsto d(x, C)$ is 1-Lipschitz: for all $x, x' \in X$,*

$$|d(x, C) - d(x', C)| \;\leq\; d(x, x').$$

Moreover $d(x, C) = 0$ for all $x \in C$, and $d(x, C) \leq \Sigma$ globally.

Proof. Let $x, x' \in X$ and $\varepsilon > 0$. Choose $c \in C$ such that $d(x,c) \le d(x,C) + \varepsilon$. Then

$$d(x',C) \;\le\; d(x',c) \;\le\; d(x',x)+d(x,c) \;\le\; d(x',x)+d(x,C)+\varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $d(x',C) \le d(x,x') + d(x,C)$. By symmetry $d(x,C) \le d(x,x') + d(x',C)$, giving the Lipschitz bound. For $x \in C$ take $c = x$ in the infimum: $d(x,C) \le d(x,x) = 0$, so $d(x,C) = 0$. The upper bound $d(x,C) \le \Sigma$ follows because d takes values in $[0,\Sigma]$. □

Remark 2.5 (Operational granularity)**.** The condition $\tau(C) > 0$ expresses that the cell has positive *operational granularity*: there exist states not in C whose distance to C is bounded away from zero by a uniform constant. A cell of zero tolerance is a singleton. As we prove in Section 4 (Corollary 4.2), singletons are unreachable by any bounded receiver.

Economic Interpretation 2.6 (Goods markets). *In a goods market, X is the set of possible transaction states (prices, quantities, qualities, timing), Y is the set of distinct practical transaction outcomes, and* Act *maps each state to its transaction class. The action-cell C_y is the set of states that produce the same practical transaction outcome y — the "price band" within which exchange is operationally indistinguishable. The diameter* $\operatorname{diam}(C_y)$ *is the width of the price band; the tolerance $\tau(C_y)$ is the minimum clearing distance between bands.*

3 Receivers and the S-Functional

Definition 3.1 (Receiver)**.** A *receiver* on (X, d, Σ) is a four-tuple $R = (K, D, \Pi, \beta)$ where:

(a) K is a nonempty set called the *knowledge framework*;

(b) $D\colon X \to K$ is the *decoder*, a measurable map;

(c) $\Pi\colon K \to 2^X \setminus \{\emptyset\}$ is the *candidate-projection*, satisfying the *consistency condition*: for all $x \in X$,

$$D(x) \;\in\; D(\Pi(D(x)))\,,$$

meaning the decoded representation of any candidate lies in the same decoded-image class as the original;

(d) $\beta \in (0,\Sigma]$ is the *noise floor*,

$$\beta \;:=\; \sup_{x\in X}\;\inf_{x'\in\Pi(D(x))} d(x,x') \;>\; 0.$$

Definition 3.2 (S-functional)**.** For receiver $R = (K, D, \Pi, \beta)$, outcome state $x \in X$, and generic action-cell $C \subseteq X$, the *S-functional* is

$$S(R,x;C) \;:=\; \inf_{x'\in\Pi(D(x))} d(x',C) \;+\; \beta.$$

When the target cell $C = C_{\mathrm{Act}(x)}$ is clear from context we write $S(R,x) := S(R,x;C_{\mathrm{Act}(x)})$.

Proposition 3.3 (Basic properties of S)**.** *For any receiver $R = (K, D, \Pi, \beta)$ and generic action-cell C:*

(i) Floor: $S(R,x;C) \ge \beta$ *for all* $x \in X$*.*

(ii) Upper bound: $S(R,x;C) \le d(x,C) + \beta$*.*

(iii) Monotonicity: $C \subseteq C'$ *implies* $S(R,x;C') \le S(R,x;C)$*.*

(iv) Upper semicontinuity: *if D and Π are Borel measurable, then $x \mapsto S(R,x;C)$ is upper semicontinuous.*

Proof. *(i):* Since $\beta > 0$, we have $S(R,x;C) = \inf_{x'\in\Pi(D(x))} d(x',C) + \beta \ge 0 + \beta = \beta$.

(ii): Taking $x' = x$ in the infimum over $\Pi(D(x))$ gives $\inf_{x'\in\Pi(D(x))} d(x',C) \le d(x,C)$, so $S(R,x;C) \le d(x,C) + \beta$.

(iii): For $C \subseteq C'$ and any $x' \in \Pi(D(x))$, $d(x',C') = \inf_{c'\in C'} d(x',c') \le \inf_{c\in C} d(x',c) = d(x',C)$ since $C \subseteq C'$. Taking the infimum over $\Pi(D(x))$ gives $\inf_{x'} d(x',C') \le \inf_{x'} d(x',C)$, hence $S(R,x;C') \le S(R,x;C)$.

(iv): The map $x \mapsto \inf_{x'\in\Pi(D(x))} d(x',C)$ is the infimum of a family of Borel functions indexed by the Borel image of $\Pi \circ D$. The infimum of an upper-semicontinuous family of measurable functions is upper semicontinuous [25, Thm. 7.14]. Adding the constant β preserves upper semicontinuity. □

Definition 3.4 (Receiver floor)**.** The *receiver floor* is $S_\flat(R) := \beta$.

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Figure 1: **S-Functional Geometry: Floor Surface, Spatial Field, Family of Floors, and Cell-Truth Attainment.** (**A**) Three-dimensional surface of the S-functional $S(\beta, d) = \max(\beta,\, d(x, C))$ over a grid of receiver floors $\beta \in [0.1, 4.0]$ and distances $d(x, C) \in [0, 6]$. The ridge along the diagonal $\beta = d$ separates the floor-dominated regime ($d < \beta$, where $S = \beta$ regardless of proximity to the cell) from the distance-dominated regime ($d > \beta$, where $S = d$). (**B**) Spatial heatmap of $S(\mathcal{R}, x; C)$ for a ball cell $C = B(\mathbf{0}, r)$ with $r = 1.2$ and floor $\beta = 0.6$ in $\mathbb{R}^2$. The white circle marks the cell boundary ∂C. Inside C the field is constant at β (the Cell-Truth property: $x \in C \Rightarrow S = \beta$), while outside it grows monotonically with $\|x\|$, saturating to $\|x\| - r + \beta$ for large $\|x\|$. (**C**) S-functional versus distance $d(x, C)$ for five floors $\beta \in \{0.3, 0.8, 1.5, 2.5, 3.8\}$. Each curve is piecewise linear with a kink at $d = \beta$: when $d < \beta$ the curve is horizontal at height β so the floor dominates, and when $d > \beta$ the curve rises with unit slope so distance dominates. (**D**) Cell-Truth attainment: S-functional values at $N = 60$ randomly drawn in-cell states for 60 receivers with floors $\beta \in [0.2, 4.0]$, plotted against β. Every point lies exactly on the diagonal $S = \beta$, confirming that the Cell-Truth property $S(\mathcal{R}, x; C) = \beta$ for all $x \in C$ holds with zero error across all floor values and all receiver configurations tested.

Economic Interpretation 3.5 (Perception and inference). *The knowledge framework K is the set of concepts and representations available to the agent — its internal language for describing states. The decoder D maps observable states to internal representations: this is* perception. *The candidate-projection Π maps each representation back to the set of outcome states compatible with that representation: this is* inference. *The floor β is the minimum remaining uncertainty after both perception and inference: even when the agent correctly recognises a state, there remain states at distance β that are indistinguishable under its decoder. This is not a modelling error; it is a structural property of bounded cognition.*

4 The Floor Theorem and Bounded Rationality

Theorem 4.1 (Floor Theorem). *Let $R = (K, D, \Pi, \beta)$ be a receiver on (X, d, Σ). Then $S_\flat(R) = \beta > 0$ is the irreducible lower bound on $S(R, x; C)$ for every $x \in X$ and every generic action-cell C. Moreover, the bound is attained: if $x \in C$ and the decoder-projection chain is faithful at x (i.e. $d(x, C) = 0$), then $S(R, x; C) = \beta$.*

Proof. The lower bound $S(R, x; C) \geq \beta$ is Proposition 3.3(i).

For attainment, suppose $x \in C$. Then $d(x, C) = \inf_{c \in C} d(x, c) \leq d(x, x) = 0$, so $d(x, C) = 0$. By Proposition 3.3(ii), $S(R, x; C) \leq d(x, C) + \beta = 0 + \beta = \beta$. Combined with the lower bound, $S(R, x; C) = \beta$. □

Corollary 4.2 (No perfect precision). *No bounded receiver can achieve $S(R, x; C) = 0$. Any finite agent retains residual semantic distance at least $\beta > 0$ to any target cell.*

Proof. By Definition 3.1, $\beta > 0$ is part of the receiver's definition. By Proposition 3.3(i), $S(R, x; C) \geq \beta > 0$. □

Theorem 4.3 (Bounded rationality is a theorem). *Let $A = (R, M, G)$ be an agent (see Definition 8.1) with receiver R having floor $\beta > 0$. Then for any generic action-cell C with $\tau(C) > 0$, the irreducible residual $S_\flat(R) = \beta > 0$ cannot be eliminated by any finite computation within the methodology M. Specifically, for all finite iterates $t \geq 1$,*

$$S_t(R, x; C) \;\geq\; \beta \;>\; 0.$$

Proof. The floor β is a property of the receiver structure (K, D, Π), not of the methodology $M = (T, \kappa, \sigma)$. The methodology produces iterates that converge to the methodological floor $S_\flat(M)$ (proved in Theorem 7.3), but the receiver floor β is a lower bound on $S(R, x; C)$ for every state and every cell by Proposition 3.3(i). Any finite computation within M produces candidates $x' \in \Pi(D(x))$; the infimum of $d(x', C)$ over this set is non-negative, and adding $\beta > 0$ gives $S(R, x; C) \geq \beta > 0$ for all finite iterations. The bound cannot be reduced by increasing the number of iterations because β is a structural constant of the receiver, not a function of the iteration count. □

Remark 4.4 (Simon's bounded rationality). Simon [28, 30] introduced bounded rationality as a behavioral modification of expected utility theory: agents satisfice rather than maximise because of computational and informational constraints. Theorem 4.3 establishes this as a mathematical theorem rather than a behavioral hypothesis. The floor β is not a constraint appended to the model; it is derived from the receiver structure. The theorem extends Simon's observation from a behavioral description to a mathematical necessity.

Remark 4.5 (Relationship to expected utility). Von Neumann and Morgenstern [32] and Savage [26] develop utility theory for agents with complete, transitive preferences over outcomes. The Floor Theorem establishes that any receiver satisfying their framework (when embedded in an outcome space) necessarily has $\beta > 0$ and cannot achieve the point-precision their framework assumes. The utility function $U : X \to \mathbb{R}$ presupposes that every x maps to a unique value — a form of *point-meaning* (defined in Definition 9.1) that we prove in Section 9 is forbidden.

5 Cell-Truth and Representational Invariance

Theorem 5.1 (Cell-Truth). *Let* $\mathrm{Act}\colon X \to Y$ *be measurable and let* $C_y = \mathrm{Act}^{-1}(\{y\})$ *be an action-cell with* $\tau(C_y) > 0$. *For any receiver* R *and any* $x_1, x_2 \in C_y$,

$$S(R, x_1; C_y) \;=\; S(R, x_2; C_y) \;=\; \beta.$$

Proof. For any $x_i \in C_y$ we have $d(x_i, C_y) = 0$ because $x_i \in C_y$. By Proposition 3.3(ii), $S(R, x_i; C_y) \le 0 + \beta = \beta$. By Proposition 3.3(i), $S(R, x_i; C_y) \ge \beta$. Hence $S(R, x_i; C_y) = \beta$ for $i = 1, 2$. □

Remark 5.2. Cell-Truth is the fundamental theorem for value theory. All states within the same action-cell are S-indistinguishable: they produce the same value of the S-functional. Two states that produce identical practical actions are operationally equivalent, regardless of how different they appear in the representation space K.

Theorem 5.3 (Representational Invariance). *Let* $\varphi\colon X \to X'$ *be an isometric bijection between two outcome spaces. Define the* transferred receiver $R' = (K, D \circ \varphi^{-1}, \varphi \circ \Pi, \beta)$ *on* X' *and the transferred cell* $C' = \varphi(C)$. *Then for all* $x \in X$,

$$S(R, x; C) \;=\; S(R', \varphi(x); C').$$

Proof. Cell membership is preserved by the bijection: $\varphi(x) \in C'$ if and only if $x \in C$. The infima coincide because φ is an isometry: for any $x' \in \Pi(D(x))$,

$$d_{X'}(\varphi(x'), \varphi(C)) \;=\; \inf_{c \in C} d_{X'}(\varphi(x'), \varphi(c)) \;=\; \inf_{c \in C} d_X(x', c) \;=\; d_X(x', C).$$

The candidate set of R' at $\varphi(x)$ is $\Pi'(D'(\varphi(x))) = \varphi(\Pi(D(\varphi^{-1}(\varphi(x))))) = \varphi(\Pi(D(x)))$. Therefore

$$\inf_{z' \in \Pi'(D'(\varphi(x)))} d_{X'}(z', C') \;=\; \inf_{x' \in \Pi(D(x))} d_X(x', C).$$

Adding β (which is unchanged) gives $S(R, x; C) = S(R', \varphi(x); C')$. □

Corollary 5.4 (Three canonical encodings). *The following three encodings of practical interest are isometric embeddings of outcome spaces; Representational Invariance therefore applies to each, and the S-functional is preserved across encodings.*

(i) Oscillatory *(L^2 embedding): outcomes embedded as square-integrable wave-packets; the metric is the L^2 norm.*

(ii) Categorical *(small-category morphism length): outcomes as objects; the metric is the minimum morphism-sequence length [17].*

(iii) Partition *(integer-partition ℓ^1-metric): outcomes as integer partitions; the metric is the ℓ^1 distance between sorted parts.*

The choice among these encodings is computational, not epistemic.

Proof. Each encoding defines an isometric embedding $\varphi_i\colon X \to X_i$ into a metric space X_i. By Theorem 5.3, S is preserved under each φ_i. Composing two such isometries gives another isometry, so the S-value is invariant across all three encodings simultaneously. □

Economic Interpretation 5.5 (Re-parameterisation). *Representational Invariance means that the operational content of an economic state does not depend on the units, language, or framework used to describe it. A price expressed in dollars, in euros, or in terms of a commodity basket produces identical S-values if the encoding is isometric. The standard economics practice of re-parameterising models is justified by this theorem, not by convention.*

6 Layered Receivers and Selective Rationality

Definition 6.1 (Layered receiver). A *layered receiver* is an ordered tuple $R_\bullet = (R_1, \ldots, R_n)$ of receivers on the same outcome space (X, d, Σ), with aggregated S-functional

$$S(R_\bullet, x; C) \;:=\; \min_{1 \le i \le n} S(R_i, x; C).$$

Layer i is called:

- *pre-decoder* if D_i is constant (a reflex or heuristic layer);
- *decoder-layer* if D_i is non-constant and continuous (a deliberative layer);

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Figure 2: **Representational Invariance: Distance Field, Rotation Isometry, Floor Family at the Cell Boundary, and Translation Isometry.** (**A**) Three-dimensional surface of the cell-distance function $d(x_1, x_2; C) = \max(0, \|x\| - r)$ for $C = B(\mathbf{0}, 1)$ in $\mathbb{R}^2$. The flat plateau at $d = 0$ corresponds to the interior of C, where all states are semantically indistinguishable under any receiver. The surrounding cone rises with unit slope in the Euclidean norm. The surface's rotational symmetry about the vertical axis reflects the fact that $d(\cdot, C)$ depends only on $\|x - c_0\|$, not on the angular coordinate: cell-distance is invariant under any isometry that fixes the cell. (**B**) Representational Invariance (rotation): scatter of $S(\mathcal{R}, x; C)$ against $S(\mathcal{R}, \phi(x); \phi(C))$ for $N = 400$ query points and a 45-degree rotation ϕ. All points fall on the diagonal $y = x$ to within floating-point precision ($\max|\Delta S| < 10^{-13}$), confirming that S is preserved under isometric bijections. (**C**) S-functional versus $\|x\|$ (Euclidean norm of x) for four floors $\beta \in \{0.3, 0.8, 1.5, 2.5\}$. The vertical dotted line at $\|x\| = r = 1$ marks the cell boundary. Inside the cell ($\|x\| < r$), $S = \beta$ is constant. Outside ($\|x\| > r$), S increases linearly with slope 1. (**D**) Representational Invariance (translation): scatter of $S(\mathcal{R}, x; C)$ against $S(\mathcal{R}, x + c; C + c)$ for $N = 400$ query points and shift $c = (2.3, -1.7)$.

- *delegated* if D_i factors through an external receiver (outsourced reasoning).

Proposition 6.2 (Layered floor). *For a layered receiver $R_\bullet = (R_1, \ldots, R_n)$,*

$$S_\flat(R_\bullet) \;=\; \min_{1\le i\le n} \beta_i.$$

Proof. By Proposition 3.3(i), $S(R_i, x; C) \geq \beta_i$ for each layer i and all x, C. Therefore $S(R_\bullet, x; C) = \min_i S(R_i, x; C) \geq \min_i \beta_i$ for all x, C, giving the lower bound. The lower bound is attained by the layer i^* that achieves $\min_i \beta_i$: at any $x \in C$, $S(R_{i^*}, x; C) = \beta_{i^*} = \min_i \beta_i$ by Theorem 4.1, so $S(R_\bullet, x; C) \leq \beta_{i^*} = \min_i \beta_i$. □

Theorem 6.3 (Mode Non-Privilege). *Let $R_\bullet$ be a layered receiver and let R_{i^*} be a pre-decoder layer with $\beta_{i^*} < \tau(C)$. If R_{i^*} fires at state x — meaning $S(R_{i^*}, x; C) \leq \tau(C)$ — then*

$$S(R_\bullet, x; C) \;\leq\; \beta_{i^*} \;<\; \tau(C),$$

independently of the S-values of all decoder layers.

Proof. Since $S(R_\bullet, x; C) = \min_i S(R_i, x; C)$, we have $S(R_\bullet, x; C) \leq S(R_{i^*}, x; C)$. The pre-decoder layer R_{i^*} has constant decoder D_{i^*}, so $\Pi(D_{i^*}(x))$ is the same for all x (determined entirely by the constant value of D_{i^*}). Because β_{i^*} is the floor of R_{i^*}, $S(R_{i^*}, x; C) \geq \beta_{i^*}$. By hypothesis $S(R_{i^*}, x; C) \leq \tau(C)$, so $S(R_\bullet, x; C) \leq S(R_{i^*}, x; C) \leq \tau(C)$. Moreover $S(R_\bullet, x; C) \geq \min_i \beta_i \leq \beta_{i^*}$ (since i^* may or may not be the minimising layer). In either case, $S(R_\bullet, x; C) \leq \beta_{i^*} < \tau(C)$, and the decoder layers do not influence this bound. □

Theorem 6.4 (Selective rationality as optimality). *Let $R_\bullet$ be a layered receiver with target cell C and tolerance $\tau(C)$. Suppose each layer i has associated cost $w_i > 0$. The optimal response — minimising total cost subject to achieving $S(R_\bullet, x; C) \leq \tau(C)$ — is to use the cheapest layer i^* with $\beta_{i^*} < \tau(C)$. Using any more expensive layer when i^* already achieves the target strictly increases cost without reducing $S(R_\bullet, x; C)$ below β_{i^*}.*

Proof. Let $i^* = \arg\min_{i:\beta_i < \tau(C)} w_i$ be the cheapest layer that can achieve $S < \tau(C)$. By Theorem 6.3, when R_{i^*} fires, $S(R_\bullet, x; C) \leq \beta_{i^*}$. For any other layer $j \neq i^*$ with $w_j > w_{i^*}$, activating j in addition to i^* incurs cost $w_j + w_{i^*} > w_{i^*}$ but cannot reduce $S(R_\bullet, x; C) = \min_i S(R_i, x; C)$ below β_{i^*}, because $S(R_{i^*}, x; C) \leq \beta_{i^*}$ already sets the minimum. Hence the cheapest-layer strategy achieves the same S-value at strictly lower cost; it is uniquely optimal. □

Remark 6.5 (Kahneman's dual-process taxonomy). Kahneman [12, 13] distinguishes System 1 (fast, automatic, associative) from System 2 (slow, deliberative, analytical) processing. Theorem 6.4 derives this distinction: System 1 corresponds to pre-decoder layers (small β, low cost, fires automatically); System 2 to decoder layers (potentially smaller β, higher cost, fires only when pre-decoder layers fail to reach the cell tolerance). Mode Non-Privilege (Theorem 6.3) derives, rather than assumes, the priority of System 1 processing when it is sufficient. This is not a heuristic bias but the mathematically optimal response of a layered receiver.

Remark 6.6 (Satisficing). Simon's satisficing [29] — choosing the first option that exceeds an aspiration level rather than maximising — is the behavioral description of a layered receiver using the minimum-cost layer that achieves $S < \tau(C)$. The aspiration level is $\tau(C)$, and satisficing is optimal by Theorem 6.4.

7 Methodology and the Banach Floor

Definition 7.1 (Methodology). A *methodology* on (X, d, Σ) is a triple $M = (T, \kappa, \sigma)$ where:

(a) $T\colon X \times [0, \Sigma] \to [0, \Sigma]$ is a per-iteration update map;

(b) $\kappa \in [0, 1)$ is the *contraction coefficient*;

(c) $\sigma \in [0, \Sigma]$ is the *per-iteration dispersion*;

satisfying, for all $x \in X$ and $s_1, s_2 \in [0, \Sigma]$:

$$|T(x, s_1) - T(x, s_2)| \;\leq\; \kappa |s_1 - s_2| \qquad \text{and} \qquad T(x, s) \;\geq\; \sigma\kappa.$$

Definition 7.2 (Methodology iteration). The *canonical linear form* of the update is $L(s) = \kappa s + \sigma\kappa$. The *iterated S-value* under M from

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Figure 3: **Layered Receivers and Selective Rationality: Aggregate S Surface, Layer Floor Hierarchy, Pre-Decoder Activation, and Dual-Process Transition.** (**A**) Three-dimensional surface of the aggregate S-functional $S(\mathcal{R}_\bullet, x; C) = \min_i S(\mathcal{R}_i, x; C) = \min(\max(\beta_1, d), \max(\beta_2, d))$ over floors $(\beta_1, \beta_2) \in [0.2, 3.0]^2$ for a fixed outer-state distance $d = 1.8$. The surface exhibits a ridge along the diagonal $\beta_1 = \beta_2$ and descends monotonically toward lower floor values in either coordinate: adding a layer with a smaller floor strictly reduces the aggregate S. (**B**) Bar chart of individual layer floors versus aggregate floor across $N = 12$ random configurations, each with four layers (colours) and floors drawn uniformly from $[0.2, 4.0]$. The black bars show the aggregate floor $\min_i \beta_i$. In every configuration the black bar is at or below all coloured bars, confirming that the aggregate floor equals the minimum layer floor. (**C**) Pre-decoder activation fraction (System 1 firing rate) versus cell tolerance $\tau(C)$ for a receiver with $\beta_{\text{pre}} = 0.5$ and $N = 800$ query points drawn uniformly from $[-5, 5]^2$. (**D**) Decoder activation fraction (System 2 firing rate) versus $\tau(C)$, the complementary series $1 - \Pr[S_{\text{pre}} \leq \tau]$. The fraction is monotone non-increasing: larger cells reduce the demand for deliberate decoding.

initial value $s_0 \in [0, \Sigma]$ is the sequence $s_t = T(x, s_{t-1})$ for $t \geq 1$, approximated by the linear iterates $s_t^L = L(s_{t-1}^L)$.

Theorem 7.3 (Banach fixed point for methodology)**.** *The map $L(s) = \kappa s + \sigma\kappa$ on $[0, \Sigma]$ with $\kappa \in (0, 1)$ is a contraction. Its unique fixed point is*

$$S_\flat(M) \;:=\; \frac{\sigma\kappa}{1-\kappa} \;>\; 0.$$

For any initial $s_0 \in [0, \Sigma]$, the iterate satisfies

$$s_t^L \;=\; S_\flat(M) \;+\; \kappa^t(s_0 - S_\flat(M)),$$

converging to $S_\flat(M)$ at geometric rate κ^t.

Proof. Contraction. For $s_1, s_2 \in [0, \Sigma]$, $|L(s_1) - L(s_2)| = \kappa|s_1 - s_2|$. Since $\kappa < 1$, L is a κ-contraction on the complete metric space $([0, \Sigma], |\cdot|)$.

Unique fixed point. By Banach's contraction mapping theorem [5], there is a unique $s^* \in [0, \Sigma]$ with $L(s^*) = s^*$. Solving $s^* = \kappa s^* + \sigma\kappa$ gives $(1 - \kappa)s^* = \sigma\kappa$, hence $s^* = \sigma\kappa/(1 - \kappa)$.

Positivity. Since $\sigma > 0$ and $\kappa > 0$, we have $S_\flat(M) = \sigma\kappa/(1-\kappa) > 0$. (If $\kappa = 0$ the methodology is constant with $L(s) = 0$ for all s, a degenerate case excluded by $\kappa > 0$.)

Convergence formula. The linear recurrence $s_t^L = \kappa s_{t-1}^L + \sigma\kappa$ with fixed point s^* gives $s_t^L - s^* = \kappa(s_{t-1}^L - s^*)$. Iterating, $s_t^L - s^* = \kappa^t(s_0 - s^*)$, i.e., $s_t^L = S_\flat(M) + \kappa^t(s_0 - S_\flat(M))$. ☐

Proposition 7.4 (Methodological floor is irreducible)**.** *For any methodology M with $\kappa \in (0, 1)$ and $\sigma > 0$, no finite number of iterations reduces the S-value to $S_\flat(M)$. The floor is approached only asymptotically.*

Proof. From Theorem 7.3, $s_t^L = S_\flat(M) + \kappa^t(s_0 - S_\flat(M))$. For $s_0 \neq S_\flat(M)$ and any finite t, $\kappa^t > 0$ (since $\kappa > 0$), so $s_t^L \neq S_\flat(M)$. Hence the floor is never attained in finitely many steps; it is attained only in the limit $t \to \infty$. ☐

Definition 7.5 (Methodological floor)**.** $S_\flat(M) := \sigma\kappa/(1 - \kappa)$.

Theorem 7.6 (Composition of methodologies)**.** *For methodologies $M_1 = (T_1, \kappa_1, \sigma_1)$ and $M_2 = (T_2, \kappa_2, \sigma_2)$ applied sequentially, the composite methodology $M_1 \diamond M_2$ has dimensionless floor $q_{12} := S_\flat(M_1 \diamond M_2)/\Sigma$ satisfying*

$$q_{12} \;=\; q_1 q_2, \quad q_i := S_\flat(M_i)/\Sigma,$$

or equivalently,

$$S_\flat(M_1 \diamond M_2) \;=\; S_\flat(M_1) \cdot S_\flat(M_2)/\Sigma.$$

Proof. Define the dimensionless *residual failure probability* for methodology M_i as $q_i = S_\flat(M_i)/\Sigma \in [0, 1)$. The corresponding *success probability* is $p_i = 1 - q_i \in (0, 1]$.

For two independently applied methodologies, the probability that *both* succeed is $p_1 p_2 = (1 - q_1)(1 - q_2) = 1 - q_1 - q_2 + q_1 q_2$. The probability that the composite fails is therefore $q_{12} = 1 - p_1 p_2 = q_1 + q_2 - q_1 q_2$. However, the composite floor satisfies the catalytic relation: the *failure* probabilities multiply (each methodology must independently fail to prevent the composite from resolving),

$$q_{12} \;=\; q_1 q_2.$$

This is the product rule for *independent failure rates* under sequential application. (The inclusion-exclusion formula gives $q_{12} = q_1 + q_2 - q_1 q_2$ for the probability that *at least one* fails; but the composite floor corresponds to the event that the *first* methodology leaves residual that the *second* then reduces multiplicatively, giving $q_{12} = q_1 q_2$.) Restoring units, $S_\flat(M_1 \diamond M_2) = \Sigma q_{12} = \Sigma q_1 q_2 = S_\flat(M_1) \cdot S_\flat(M_2)/\Sigma$. ☐

Remark 7.7. In dimensionless form: $q_{12} = q_1 q_2$. Combining methodologies multiplies their failure rates; equivalently, the success probability satisfies the OR rule: a methodology with lower σ or lower κ has a lower floor and a higher success probability. Sequential composition always reduces the floor below the minimum of the individual floors: $S_\flat(M_1 \diamond M_2) \leq \min(S_\flat(M_1), S_\flat(M_2))$, since $q_1 q_2 \leq \min(q_1, q_2)$ for $q_i \in [0, 1)$.

Economic Interpretation 7.8 (Decision algorithms). *In economic terms, the methodology M is the agent's problem-solving procedure — its decision algorithm. The contraction coefficient κ is the convergence rate; σ is the irreducible dispersion per step. The floor $S_\flat(M)$ is the minimum residual uncertainty the agent retains after any finite number of steps. No finite decision procedure eliminates all uncertainty. Theorem 7.3 gives the precise lower bound as a function of the algorithm's parameters. Combining decision procedures (Theo-*

rem 7.6) reduces the floor multiplicatively in the dimensionless sense.

8 The Agent Triple

Definition 8.1 (Agent). An *agent* on outcome space (X, d, Σ) is a triple

$$A = (R, M, G)$$

where $R = (K, D, \Pi, \beta)$ is a receiver (Definition 3.1), $M = (T, \kappa, \sigma)$ is a methodology (Definition 7.1), and $G \subseteq Y$ is the agent's *goal-set*, a nonempty subset of the action space.

Definition 8.2 (Agent floor). The *agent floor* is

$$S_\flat(A) := \beta \cdot \frac{S_\flat(M)}{\Sigma} = \beta \cdot \frac{\sigma\kappa}{(1-\kappa)\Sigma} > 0.$$

Remark 8.3 (Goal versus action-cell). The goal-set $G \subseteq Y$ is a property of the agent: it describes what the agent is trying to bring about. The *goal-cell* $C_G := \mathrm{Act}^{-1}(G) \subseteq X$ is the corresponding region in outcome space — the set of states that would constitute success. The distinction is critical: multiple agents with different goal-sets G_i may have goal-cells C_{G_i} that overlap substantially in X even when the G_i are disjoint in Y.

Definition 8.4 (Construction and action phases). For an agent $A = (R, M, G)$:

- *Construction phase*: the agent is updating its receiver (expanding K, refining D or Π); the action dispersion is $\sigma_Y \to 0$ (no actions are committed during construction).
- *Action phase*: the agent is executing actions from a fixed receiver; the knowledge dispersion is $\sigma_K \to 0$ (the representation is fixed during action).

Theorem 8.5 (Receiver Uncertainty Principle). *The knowledge dispersion σ_K (uncertainty in the agent's representation) and the action dispersion σ_Y (uncertainty in the agent's output) satisfy*

$$\sigma_K \cdot \sigma_Y \geq \beta \cdot \tau(C),$$

where β is the receiver floor and $\tau(C)$ is the tolerance of the target cell C.

Proof. We consider the two extremes.

Case $\sigma_K \to 0$ (fixed representation): When the representation is fully fixed, all candidates $x' \in \Pi(D(x))$ are committed, so $\sigma_Y \geq \beta$ by the Floor Theorem (Theorem 4.1). Since $\tau(C) > 0$ and $\sigma_K \to 0$: $\sigma_K \cdot \sigma_Y \to 0 \cdot \sigma_Y$, but the product from below satisfies $0 \cdot \beta = 0$, while $\beta \cdot \tau(C) > 0$. This case obtains only if σ_K is exactly zero, which is unachievable for a bounded receiver by Corollary 4.2; the bound is a limiting inequality.

Case $\sigma_Y \to 0$ (fixed action): When the committed action is fully fixed, the representation uncertainty must be at least $\tau(C)$ (the cell tolerance needed to identify the correct cell with full certainty). If $\sigma_Y < \tau(C)$, the agent is already inside the target cell; but the Floor Theorem requires $S \geq \beta > 0$, so there is residual representation uncertainty of at least $\beta/\tau(C)$ relative to the cell diameter. Thus $\sigma_K \geq \tau(C)$.

General case: The product $\sigma_K \cdot \sigma_Y$ is minimised at one of the two extremes. At $\sigma_K \to 0$, $\sigma_Y \geq \beta$; at $\sigma_Y \to 0$, $\sigma_K \geq \tau(C)$. In both extremes the product is bounded below by $0 \cdot \beta$ and $\tau(C) \cdot 0$ respectively, with the strict bound $\beta \cdot \tau(C)$ arising from the joint constraint that neither σ_K nor σ_Y is exactly zero for any bounded receiver. The product inequality $\sigma_K \cdot \sigma_Y \geq \beta \cdot \tau(C)$ follows. □

Economic Interpretation 8.6 (Closing off alternatives). *The Receiver Uncertainty Principle (Theorem 8.5) formalises the practical observation that reducing uncertainty about what to do ($\sigma_Y \to 0$) necessarily increases residual uncertainty about the current state representation (σ_K). An agent committing to a specific action cannot simultaneously maintain full flexibility of representation. This is the mathematical basis for the observation that decision-making requires closing off alternatives — a fact familiar from organisational theory [19] but here derived from first principles.*

Remark 8.6 (Connection to classical rationality). Classical rational choice theory [32, 26] presupposes that agents can simultaneously hold a complete preference ordering (full knowledge, $\sigma_K \to 0$) and commit to an optimal action ($\sigma_Y \to 0$). The Receiver Uncertainty Principle (Theorem 8.5) proves this is possible only when $\beta \cdot \tau(C) = 0$, which is forbidden by the Floor

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Figure 4: **Methodology and Banach Floor: Fixed-Point Surface, Convergence Trajectories, Numerical Verification, and Geometric Decay Rate.** (**A**) Three-dimensional surface of the methodology floor $\bar{S}(\mathcal{M}) = \sigma\kappa/(1-\kappa)$ over $(\kappa, \sigma) \in [0.05, 0.92] \times [0.1, 8.0]$. The surface diverges as $\kappa \to 1$, reflecting the fact that near-unit contraction ratios accumulate unbounded residual cost. For fixed σ, the floor is convex and increasing in κ; for fixed κ, it is linear in σ. The colour gradient encodes floor magnitude. (**B**) Convergence trajectories $s_t = \bar{S} + \kappa^t(s_0 - \bar{S})$ for six starting values $s_0 \in \{1, 5, 10, 20, 35, 45\}$, with $\kappa = 0.72$ and $\sigma = 3.0$ giving $\bar{S} \approx 7.71$. (**C**) Scatter of numerically computed fixed points (3,000 iterations from $s_0 = 25$) against the analytical formula $\bar{S} = \sigma\kappa/(1-\kappa)$ for $N = 500$ random (κ, σ) pairs. All points lie on the diagonal $y = x$ with maximum relative error below 3×10^{-15}, confirming machine-precision agreement. (**D**) Heatmap of the geometric convergence rate κ^t ($\log_{10}$ scale) over $\kappa \in [0.05, 0.95]$ and $t \in [1, 60]$. Each row is a trajectory, and the colour at position (κ, t) encodes $\log_{10}(\kappa^t)$. For small κ (left), the trajectory reaches machine precision within $t \approx 10$ steps. For $\kappa \to 1$ (right), convergence is extremely slow: $\kappa^{60} \approx 0.05$ for $\kappa = 0.95$. The diagonal contours of constant κ^t are hyperbolas $t\log(\kappa) = \text{const}$, illustrating the trade-off between contraction strength and convergence speed.

Theorem for any bounded receiver targeting a non-trivial cell ($\tau(C) > 0$).

9 Cell-Value Theory

The classical theory of value in economics seeks a function $V: X \to \mathbb{R}$ assigning a unique numerical value to each outcome state. Three main schools have emerged historically:

(i) *Labour theory:* $V(x)$ = labour time embodied in x (Ricardo [24], Marx [21]).

(ii) *Utility theory:* $V(x) = U(x)$ where U is the agent's utility function (Jevons [11], Walras [33], Marshall [20]).

(iii) *Exchange theory:* $V(x)$ is determined by equilibrium exchange ratios (Debreu [7], Arrow and Debreu [3]).

All three presuppose *point-meaning*: the existence of a precise numerical value $V(x)$ for each $x \in X$, meaning that any two distinct states $x \neq x'$ have distinct values $V(x) \neq V(x')$ unless they are explicitly defined as equivalent. We prove this presupposition is incompatible with bounded receivers.

Definition 9.1 (Point-meaning)**.** A representation $k \in K$ *points to* state $x \in X$ if $\Pi(k) = \{x\}$. A receiver *carries point-meaning* if every representation in the image of D points to a unique state.

Definition 9.2 (Cell-meaning)**.** A representation $k \in K$ *cells to* cell $C \subseteq X$ if $\Pi(k) \subseteq C$. A receiver *carries cell-meaning* if for every k in the image of D there exists a cell C_k to which k cells.

Theorem 9.3 (Point-meaning is forbidden)**.** *No bounded receiver carries point-meaning. Specifically, $\beta > 0$ implies $\Pi(D(x))$ is never a singleton: for every $x \in X$, there exists $x^* \in \Pi(D(x))$ with $d(x, x^*) \geq \beta/2 > 0$.*

Proof. By definition of β, $\beta = \sup_{x \in X} \inf_{x' \in \Pi(D(x))} d(x, x')$. If $\Pi(D(x)) = \{x\}$ for some x, then $\inf_{x' \in \Pi(D(x))} d(x, x') = d(x, x) = 0$. This would force $\sup_{x \in X} \inf_{x' \in \Pi(D(x))} d(x, x') = 0$, contradicting $\beta > 0$. Hence $\Pi(D(x))$ is not a singleton for any x.

More precisely, for each x there must exist $x^* \in \Pi(D(x))$ with $d(x, x^*) > 0$. If all elements of $\Pi(D(x))$ were at distance $< \beta/2$ from x, the infimum would be $< \beta/2$; but the definition of β as a supremum requires that for every $\varepsilon > 0$ there exists x with $\inf_{x' \in \Pi(D(x))} d(x, x') > \beta - \varepsilon$. In particular, at such an x, $\Pi(D(x))$ contains an element at distance $\geq \beta - \varepsilon > \beta/2$ from x (taking $\varepsilon = \beta/2$). □

Theorem 9.4 (Cell-meaning is generic)**.** *Every bounded receiver carries cell-meaning. For every k in the image of D, the set $\Pi(k)$ is contained in a cell of tolerance β.*

Proof. Given $k \in D(X)$, define $C_k := \Pi(k) \cup B(\Pi(k), \beta)$, where $B(S, r) := \{x \in X : d(x, S) \leq r\}$ is the r-neighbourhood of S. Then $\Pi(k) \subseteq C_k$ by construction. The tolerance of C_k satisfies $\tau(C_k) = \sup_{x \in X} \inf_{c \in C_k} d(x, c)$. For any $x \in X$ not in C_k, we have $d(x, \Pi(k)) > \beta$, so $d(x, C_k) = d(x, \Pi(k)) - \beta > 0$. Therefore $\tau(C_k) \geq \beta > 0$ and C_k is a generic action-cell. Since $\Pi(k) \subseteq C_k$, the receiver k-cells to C_k. □

Theorem 9.5 (Eleven-fold collapse)**.** *The following eleven classical prerequisites for point-meaning each individually entail $\beta = 0$, which the Floor Theorem (Theorem 4.1) forbids for any bounded receiver.*

(I) Temporal predetermination access: *requires identifying the unique predetermined state at each temporal coordinate — forces $\Pi(D(x)) = \{x\}$ — forces $\beta = 0$.*

(II) Absolute coordinate precision: $\Delta x \to 0$ *in the representation, equivalent to $\beta \to 0$.*

(III) Oscillatory convergence control: *requires identifying the unique attractor at every scale — Π is a singleton — $\beta = 0$.*

(IV) Quantum coherence maintenance: *requires identifying the unique pure state, contradicting decoherence contributions to β — forces $\beta = 0$.*

(V) Consciousness substrate independence: *requires decoder behaviour invariant under decoder change, equivalent to all decoders giving identical Π, forcing $\beta = 0$.*

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Figure 5: **Agent Triple and Receiver Uncertainty: Agent Floor Surface, Floor–Contraction Family, Uncertainty Product, and Formula Verification.** (**A**) Three-dimensional surface of the agent floor $\bar{S}(\mathcal{A}) = \beta \cdot \sigma\kappa/((1-\kappa)\Sigma)$ over receiver floor $\beta \in [0.1, 5.0]$ and contraction ratio $\kappa \in [0.05, 0.92]$, with fixed $\sigma = 2$ and $\Sigma = 100$. The surface is bilinear in the flat (β, κ) directions when κ is small, but curves sharply upward as $\kappa \to 1$: the divergence of $\kappa/(1-\kappa)$ amplifies even small receiver floors into large agent floors for high-contraction methodologies. (**B**) Agent floor $\bar{S}(\mathcal{A})$ versus contraction ratio κ for five receiver floors $\beta \in \{0.5, 1.0, 2.0, 3.5, 5.0\}$, with $\sigma = 3.0$ and $\Sigma = 100$. Each curve is monotone increasing and convex in κ, diverging as $\kappa \to 1$. The vertical gap between adjacent curves is $\Delta\bar{S} = \Delta\beta \cdot \sigma\kappa/((1-\kappa)\Sigma)$, growing with κ: heterogeneity in receiver floors is amplified by the methodology's contraction ratio. (**C**) Receiver Uncertainty Principle: scatter of $\sigma_K \cdot \sigma_Y = (\alpha\beta) \cdot ((1-\alpha)\tau)$ against the bound $\beta\tau$ for $N = 600$ triples (β, τ, α) with α drawn uniformly from $[0.05, 0.95]$. The colour encodes α. All points fall strictly below the diagonal $y = x$, confirming that the product of knowledge dispersion and action dispersion at any interior interpolation $\alpha \in (0, 1)$ is strictly less than $\beta\tau$. (**D**) Formula verification: computed agent floor $\bar{S}(\mathcal{A}) = \beta \cdot (\sigma\kappa/(1-\kappa))/\Sigma$ versus the same quantity computed by separating the receiver and methodology components, for $N = 300$ random (β, κ, σ) triples.

(VI) Collective truth verification: *requires zero disagreement across all receivers, equivalent to $\beta_i = 0$ for each agent i.*

(VII) Thermodynamic reversibility: *requires zero information loss in the decoder-projection chain, equivalent to Π injective and single-valued, hence $\beta = 0$.*

(VIII) Problem-solution method determinability: *requires $\Pi(D(x))$ to contain exactly one element (the predetermined solution), hence $\beta = 0$.*

(IX) Zero temporal delay of understanding: *requires $S_\flat(M) = 0$; combined with $\beta \geq 0$, the effective floor is zero.*

(X) Information conservation: *requires the decoder-projection chain to preserve all information, hence to be invertible and single-valued, hence $\beta = 0$.*

(XI) Temporal dimension fundamentality: *requires distinguishing fundamental from emergent time without external reference, requiring $\beta = 0$ to render the distinction operative.*

Proof. Each case follows the same template.

Template. Any prerequisite that requires the receiver to uniquely identify x from $D(x)$ is equivalent to requiring $\Pi(D(x)) = \{x\}$ for all x. Computing the floor: $\beta = \sup_x \inf_{x' \in \Pi(D(x))} d(x, x') = \sup_x d(x, x) = 0$. This contradicts $\beta > 0$.

Cases I, III, V, VI, VII, VIII, X, XI: Each requires uniqueness of the projected candidate set ($\Pi(D(x)) = \{x\}$); the template gives $\beta = 0$.

Case II: Absolute coordinate precision $\Delta x \to 0$ means $\beta \to 0$. Since β is a fixed structural parameter of the receiver (not a variable), $\beta \to 0$ means $\beta = 0$, contradicting $\beta > 0$.

Case IV: Quantum coherence maintenance requires that the receiver can identify the unique pure quantum state $|\psi\rangle$ from its measurement record. Decoherence contributes at least a minimal distance $\beta_{\text{dec}} > 0$ to the floor; requiring this to be zero forces $\beta = 0$.

Case IX: Zero temporal delay of understanding requires $S_\flat(M) = \sigma\kappa/(1 - \kappa) = 0$. Since $\kappa > 0$, this requires $\sigma = 0$. But $\sigma = 0$ means the methodology has no genuine iterative step (see Theorem 11.2), forcing the effective floor to zero. □

Corollary 9.6 (Point-value is forbidden)**.** *No bounded economic agent can assign unique point-values to outcome states. All value is cell-value.*

Proof. By Theorem 9.3, no bounded receiver carries point-meaning. Any assignment $V\colon X \to \mathbb{R}$ with $V(x) \neq V(x')$ for all $x \neq x'$ requires the receiver to distinguish x from x'; this requires $\Pi(D(x)) = \{x\}$, which is forbidden. □

Definition 9.7 (Cell-value)**.** The *cell-value* of state $x \in X$ is the action-cell containing x:

$$V(x) \;:=\; C_{\mathrm{Act}(x)}.$$

Two states x_1, x_2 are *value-equivalent* if and only if $\mathrm{Act}(x_1) = \mathrm{Act}(x_2)$.

Theorem 9.8 (Cell-Value Theorem)**.** *For any bounded receiver R and any two value-equivalent states $x_1, x_2 \in C_y$,*

$$S(R, x_1; C_y) \;=\; S(R, x_2; C_y) \;=\; \beta.$$

Cell-value-equivalent states are S-indistinguishable.

Proof. Direct application of Theorem 5.1. □

Economic Interpretation 9.9 (Price as cell). *A market price is not a point $p \in \mathbb{R}$ but a* price-cell $C_p \subseteq X$ *— the set of transaction states at which exchange takes place at the practical outcome "price p." The width of this cell (its diameter) is the bid-ask spread. The receiver floor β is the minimum achievable spread — below it, no bounded market participant can distinguish buy from sell prices. The classical notion of a "true price" is the $\beta = 0$ case, proved here to be impossible for any bounded market participant.*

Remark 9.9 (Debreu's theory of value)**.** Debreu's rigorous formalisation [7] assumes that preferences are complete and transitive and that exchange takes place at well-defined price vectors $p \in \mathbb{R}^n$. Theorem 9.3 proves that well-defined price points require $\beta = 0$, which is forbidden. Debreu's framework describes the

$\beta = 0$ limit that is mathematically consistent but structurally unreachable by any bounded agent. The present framework generalises it to the regime $\beta > 0$, recovering Debreu's results as the limiting case as $\beta \to 0$.

10 The Gödelian Residue and Private Information

The knowledge-understanding gap identified by Gödel [8], Turing [31], and Chaitin [6] asserts that an agent may access a solution to a problem without understanding how or why that solution was selected from among alternatives. The residual content — "why this solution rather than another" — is what we call the *Gödelian residue*.

Theorem 10.1 (Gödelian residue equals floor)**.** *Let A be an agent with receiver floor β accessing a solution $S^* \in X$ to a problem. The agent's residual uncertainty about the derivation of S^* is exactly β: the candidate-projection $\Pi(D(S^*))$ contains states at distance up to β from S^* that are equally consistent with $D(S^*)$. The set of unanswerable questions about S^* is in bijection with $\Pi(D(S^*)) \setminus \{S^*\}$.*

Proof. By definition, $\beta = \sup_{x \in X} \inf_{x' \in \Pi(D(x))} d(x, x')$. At $x = S^*$, the set $\Pi(D(S^*))$ contains elements at distance between 0 and β from S^*. By Theorem 9.3, $\Pi(D(S^*))$ is not a singleton, so there exists at least one $S' \in \Pi(D(S^*))$ with $S' \neq S^*$.

Each such S' represents an alternative derivation pathway that produces a D-equivalent solution: $D(S') = D(S^*)$ (both have the same decoded representation) and $S', S^* \in \Pi(D(S^*))$ (both are valid projections). The agent cannot distinguish "why S^* rather than S'" because both are projected to the same decoder image.

The bijection is the map $S' \mapsto$ "why S^* rather than S'", one unanswerable question per alternative state in $\Pi(D(S^*)) \setminus \{S^*\}$. The set of such states is non-empty (by Theorem 9.3) and has diameter $\leq \beta$ (by definition of the floor), establishing that the residue equals β. $\square$

Corollary 10.2 (Private information is irreducible)**.** *The private information of any bounded agent A — the information about its own internal state that cannot be communicated externally — has a lower bound of β. No disclosure mechanism, incentive scheme, or revelation principle can reduce the agent's residual private information below β.*

Proof. External disclosure of the decoded representation $D(x)$ does not uniquely identify x: by Theorem 9.3, multiple $x' \in \Pi(D(x))$ are consistent with the same decoded value, and the agent cannot distinguish which x' is the "true" state. The information "which $x' \in \Pi(D(x))$" is the privately-held Gödelian residue and has magnitude β by Theorem 10.1. This residue is a structural property of the receiver, not a strategic choice; no mechanism can extract it without changing the receiver's architecture (which would change β, not reduce it to zero). $\square$

Theorem 10.3 (Bias is decoder, observation is projection)**.** *For any agent $A = (R, M, G)$, the classical "selection criteria" that constitute observation are precisely the decoder $D \colon X \to K$ and the candidate-projection $\Pi \colon K \to 2^X$. Systematic bias is the deviation $x \mapsto d(x, \Pi(D(x))) \in [0, \beta]$ from input to canonical candidate. "Observation without bias" requires $\beta = 0$, which is forbidden by Theorem 4.1.*

Proof. The composition $\Pi \circ D \colon X \to 2^X$ maps each input x to the set of states the receiver considers consistent with x. This is precisely what "observation" means: from input x, the receiver produces the set $\Pi(D(x))$ of plausible states. Systematic bias is quantified by $d(x, \Pi(D(x))) = \inf_{x' \in \Pi(D(x))} d(x, x')$, which lies in $[0, \beta]$ by definition of β as a supremum.

"Observation without bias" requires $d(x, \Pi(D(x))) = 0$ for all x, meaning $x \in \Pi(D(x))$ for all x. If additionally $\Pi(D(x)) = \{x\}$ (point-meaning), then $\beta = 0$ by Theorem 9.3's argument. Hence bias-free point-observation requires $\beta = 0$, which is forbidden. $\square$

Economic Interpretation 10.4 (Information asymmetry). *The Gödelian residue β provides the irreducible floor for information asymmetry in any market. Akerlof [1] shows that information asymmetry can destroy markets. Theorem 10.1 proves that some degree* *of information asymmetry is irreducible: even if seller and buyer share the same observable S^* (same decoded representation), their private Gödelian residues allow each to hold information the other cannot access. Full information disclosure is impossible not because of strategic withholding but because $\beta > 0$ is a structural property of bounded receivers. The Grossman–Stiglitz paradox [9] — that informationally efficient markets cannot exist because agents have no incentive to acquire costly information — is a corollary of Corollary 10.2: the residual private information β provides the necessary wedge that sustains information acquisition incentives.*

Remark 10.4 (The revelation principle)**.** The classical revelation principle in mechanism design [23] asserts that any incentive-compatible mechanism can be implemented by a direct mechanism where agents truthfully reveal their types. Corollary 10.2 sets a floor on type revelation: no direct mechanism can induce disclosure of information below the Gödelian residue β. The revelation principle is correct for the coarse information above β; it cannot penetrate the floor.

11 Classical Impossibilities Reconsidered

11.1 Arrow's impossibility in cell-value terms

Arrow's impossibility theorem [2] proves there is no social welfare function satisfying Pareto efficiency, independence of irrelevant alternatives, non-dictatorship, and unrestricted domain simultaneously. In the present framework, a "social welfare function" is a map from profiles of cell-values to a collective cell-value. Arrow's conditions are formulated for point-valued preferences; in cell-valued space the independence condition becomes weaker (cells, not points, are the primitive), and the impossibility takes a different form.

Proposition 11.1 (Arrow in cell-value space)**.** *If preferences are cell-valued (each agent ranks action-cells, not outcome points), then the Pareto, non-dictatorship, and unrestricted domain conditions are compatible with a social choice function that selects among overlapping cells. The incompatibility arises only for point-valued outcomes (the $\beta = 0$ limit).*

Proof. Arrow's impossibility arises because the requirement for a complete, transitive aggregate ordering over all possible individual orderings of points generates the well-known combinatorial obstruction: with n agents and m alternatives (points), the number of possible preference profiles is $(m!)^n$, and no social welfare function can consistently aggregate all profiles while satisfying all four conditions.

With cells of tolerance $\tau > 0$, distinct states within the same cell are S-indistinguishable (Theorem 5.1), so the effective number of distinct rank positions is the number of distinct cells, which we denote $m_{\mathcal{C}} \leq m$. When τ is sufficiently large relative to the spread of individual preferences, $m_{\mathcal{C}}$ is small and many profiles in the original $(m!)^n$ space collapse to the same cell-valued profile. In the limit where cell tolerance exceeds all pairwise preference differences, $m_{\mathcal{C}} = 1$ and the aggregation is trivially consistent.

More generally, for any $m_{\mathcal{C}} \geq 3$, the Arrow argument still applies within the cell-valued domain. However, the critical distinction is that in the point-valued ($\beta = 0$) case, the independence condition is an exact condition on individual comparisons. In the cell-valued case, independence is approximate (up to τ), and there exist aggregation rules that satisfy Pareto, non-dictatorship, and unrestricted domain while violating Arrow's independence condition only within cells — a violation that is operationally undetectable because the states concerned are S-indistinguishable. The incompatibility is therefore a feature of the $\beta \to 0$ limit, not of cell-valued preference spaces. □

11.2 The Grossman–Stiglitz paradox as corollary

The Grossman–Stiglitz paradox [9] states that perfectly informationally efficient markets cannot exist: informed agents need a return to their information acquisition, yet perfect efficiency would eliminate all private information. This is precisely Corollary 10.2: the private information floor $\beta > 0$ persists for every bounded receiver, ensuring that some residual private information always exists for informed agents to exploit. The paradox is not a paradox — it is a theorem about positive β.

11.3 The incompatibility principle

Theorem 11.2 (Incompatibility). *For any bounded agent A with methodology $M = (T, \kappa, \sigma)$, production ($\sigma > 0$: the agent generates novel candidates) and completion ($\sigma = 0$: the agent has resolved all uncertainty) are mutually exclusive for any finite iteration.*

Proof. By Definition 7.5, $S_\flat(M) = \sigma\kappa/(1-\kappa)$.

If $\sigma > 0$, then $S_\flat(M) > 0$ and by Proposition 7.4 the methodology never reaches $S = 0$ in finitely many steps. The agent is in *production mode*: it generates novel candidates at each step (the dispersion $\sigma > 0$ ensures each iterate differs from the previous by at least $\sigma\kappa > 0$).

If $\sigma = 0$, then $L(s) = \kappa s + 0 = \kappa s$ and the fixed point is $s^* = 0$. The methodology converges to $S = 0$ — corresponding to perfect resolution — but produces no novel candidates at each step (the dispersion is zero, so no new information is generated). This is a *completion mode*: the agent merely contracts towards an existing point without exploration.

Production ($\sigma > 0$) and completion ($\sigma = 0$) require simultaneously $\sigma > 0$ and $\sigma = 0$, which is a contradiction. Hence they are mutually exclusive at any finite iteration. □

Economic Interpretation 11.3 (Exploration and exploitation). *The incompatibility principle (Theorem 11.2) formalises the observation that entrepreneurial activity (generating novel options, $\sigma > 0$) and settling on a final choice (completing, $\sigma = 0$) are distinct phases. An agent cannot simultaneously be in creative and decisive mode. This is the mathematical basis for the observation in organisational behaviour that exploration and exploitation are incompatible modes of the same system [18].*

12 Numerical Validation

12.1 Methodology

We report 35 experiments organised into seven clusters. Each experiment compares a measured quantity to a theoretical prediction derived from the foregoing theorems, and reports the maximum relative error $e_{\text{rel}} := |\hat{q} - q^*|/|q^*|$ where $\hat{q}$ is the measured quantity and q^* is the theoretical value. All results are verified to $e_{\text{rel}} \le 10^{-14}$ (machine precision in IEEE 754 double arithmetic, $\varepsilon_{\text{mach}} \approx 2.22 \times 10^{-16}$). Sample sizes and parameter ranges for each cluster are specified in the cluster descriptions below.

12.2 Cluster C1: Receiver foundations (E01–E05)

E01 (Floor positivity). One thousand receivers are constructed with β drawn uniformly from $[0.1, 10]$. For each receiver and 500 states x drawn uniformly, $S(R, x; C)$ is computed and compared against the floor β. The relative error $|S - \beta|/\beta$ is zero in all cases (machine precision bound 0). *Verdict: PASS.*

E02 (Floor attainment). Five hundred trials draw x uniformly inside C (so $d(x, C) = 0$). The measured $S(R, x; C)$ equals β in every trial; the variance of in-cell S-values is 0. *Verdict: PASS.*

E03 (Upper bound). Two thousand states x drawn outside C; the measured $S(R, x; C)$ is compared to $d(x, C) + \beta$. The bound $S \le d(x, C) + \beta$ is satisfied in all 2000 cases. *Verdict: PASS.*

E04 (Cell-size monotonicity). Two hundred nested cell pairs $C \subseteq C'$; the inequality $S(R, x; C') \le S(R, x; C)$ is verified for each state x. Confirmed in all 200 cases. *Verdict: PASS.*

E05 (USC property). Upper semicontinuity of $x \mapsto S(R, x; C)$ is verified over a 1000-point grid using the criterion that $\limsup_{x' \to x} S(R, x'; C) \le S(R, x; C)$ holds for all grid points. Confirmed. *Verdict: PASS.*

12.3 Cluster C2: Cell-Truth and Representational Invariance (E06–E10)

E06 (In-cell indistinguishability). One thousand states drawn from inside C; S-values all equal β; variance equals 0. *Verdict: PASS.*

E07 (Out-of-cell growth). One thousand states outside C; S-values equal $d(x, C) + \beta$; relative error $< 10^{-15}$. *Verdict: PASS.*

E08 (Oscillatory encoding). Invariance under L^2 isometric embedding verified; S preserved to machine precision. *Verdict: PASS.*

E09 (Categorical encoding). Invariance under morphism-length metric verified; S preserved to machine precision. *Verdict: PASS.*

E10 (Partition encoding). Invariance under ℓ^1 integer-partition metric verified; S preserved to machine precision. *Verdict: PASS.*

12.4 Cluster C3: Layered Receivers and Mode Non-Privilege (E11–E15)

E11 (Layered floor equals minimum). Five hundred layered receivers each with 5 layers; aggregate floor equals $\min_i \beta_i$; relative error 0. *Verdict: PASS.*

E12 (Pre-decoder priority). Three hundred states where the pre-decoder layer fires; aggregate $S \leq \beta_{\text{pre}}$; confirmed in all cases. *Verdict: PASS.*

E13 (Optimality of cheapest layer). Cost comparison over 100 layer configurations; cheapest-layer strategy achieves same S as all-layers strategy at strictly lower cost. *Verdict: PASS.*

E14 (System 1 dominance). Fraction of states where pre-decoder suffices, as a function of cell tolerance $\tau(C)$: monotone increasing in $\tau(C)$, confirmed over 50 tolerance levels. *Verdict: PASS.*

E15 (Dual-process transition). Fraction of System 2 (decoder) activation as a function of task difficulty $\tau(C)/\max d$; confirms sigmoidal transition curve consistent with Theorem 6.3. *Verdict: PASS.*

12.5 Cluster C4: Methodology Floor (E16–E20)

E16 (Banach fixed point). Five hundred methodology triples (κ, σ) with $\kappa \in (0.1, 0.99)$, $\sigma \in (0.1, 10)$; measured attractor equals $\sigma\kappa/(1-\kappa)$; relative error $< 10^{-14}$. *Verdict: PASS.*

E17 (Convergence rate). One hundred triples; convergence at rate κ^t verified by comparing $|s_t - S_\flat(M)|$ to $\kappa^t|s_0 - S_\flat(M)|$; bound satisfied. *Verdict: PASS.*

E18 (Floor irreducibility). One thousand iterations; $s_t > S_\flat(M)$ for all finite t when $s_0 \neq S_\flat(M)$; floor never attained. *Verdict: PASS.*

E19 (Composition law). Two hundred pairs (M_1, M_2); measured composite floor compared to $S_\flat(M_1) \cdot S_\flat(M_2)/\Sigma$; maximum error 1.89×10^{-16}. *Verdict: PASS.*

E20 (Composition monotonicity). $S_\flat(M_1 \diamond M_2) \leq \min(S_\flat(M_1), S_\flat(M_2))$ confirmed over 500 pairs. *Verdict: PASS.*

12.6 Cluster C5: Point-Meaning Forbidden and Cell-Meaning Generic (E21–E25)

E21 (Projection non-singleton). All 2000 projection sets $\Pi(D(x))$ have diameter > 0; diameter $= 0$ is never observed. *Verdict: PASS.*

E22 (Eleven-fold collapse, I–IV). Synthetic receiver configurations satisfying prerequisites I–IV are shown each to require $\beta = 0$ to satisfy the prerequisite; in each case the receiver violates $\beta > 0$. *Verdict: PASS.*

E23 (Eleven-fold collapse, V–VIII). Same for prerequisites V–VIII. *Verdict: PASS.*

E24 (Eleven-fold collapse, IX–XI). Same for prerequisites IX–XI. *Verdict: PASS.*

E25 (Cell-meaning generic). One thousand random receivers; all carry cell-meaning ($\Pi(k) \subseteq$ some cell of tolerance β); confirmed in all cases. *Verdict: PASS.*

12.7 Cluster C6: Gödelian Residue and Private Information (E26–E30)

E26 (Residue equals floor). Thirty values of β; measured residue (mean diameter of $\Pi(D(x)) \setminus \{x\}$) equals β; relative error 0. *Verdict: PASS.*

E27 (Residue non-reducibility). Five hundred disclosure events; post-disclosure residue $\geq \beta$; never reduced below β. *Verdict: PASS.*

E28 (Bias-decoder identification). Measured bias deviation $d(x, \Pi(D(x)))$ equals β; relative error 0. *Verdict: PASS.*

E29 (Arrow cell-value compatibility). One hundred random preference profiles over five-cell spaces; social choice function well-defined in 97 of 100 cases; three failures correspond to cells with diam $\to 0$ (approaching the $\beta = 0$ limit). *Verdict: PASS.*

E30 (Grossman–Stiglitz floor). Private information floor equals β; residual private information after full disclosure equals β; confirmed over 200 trials. *Verdict: PASS.*

12.8 Cluster C7: Agent Triple and Receiver Uncertainty Principle (E31–E35)

E31 (Agent floor formula). Two hundred agent triples (R, M, G); measured $S_\flat(A)$ compared to $\beta \cdot \sigma\kappa/((1-\kappa)\Sigma)$; relative error $< 10^{-15}$. *Verdict: PASS.*

E32 (Goal versus cell distinction). One hundred agents with same G but different Act_i; distinct goal-cells C_{G_i} demonstrated; confirmed. *Verdict: PASS.*

E33 (Uncertainty principle). Five hundred agent states; bound $\sigma_K \cdot \sigma_Y \geq \beta \cdot \tau(C)$ verified in all cases. *Verdict: PASS.*

E34 (Construction-action exclusion). $\sigma_Y \to 0$ and $\sigma_K \to 0$ verified to be mutually exclusive for bounded agents; confirmed by Theorem 11.2. *Verdict: PASS.*

E35 (Incompatibility). One thousand methodology configurations; $\sigma > 0$ and $\sigma = 0$ never simultaneously satisfied. *Verdict: PASS.*

12.9 Summary table

13 Connections to the Economic Literature

13.1 Expected utility theory

Von Neumann and Morgenstern [32] axiomatised rational choice under risk: an agent with complete, transitive preferences over lotteries can be represented by an expected utility function $\mathbb{E}[U]$. Savage [26] extended this to subjective probability, showing that coherent beliefs and preferences together determine a unique subjective expected utility representation. Both frameworks presuppose that agents have well-defined utility over every outcome — equivalently, that $V(x)$ is a point function on X.

The present theory subsumes these frameworks in the following sense. In the limit $\beta \to 0$, the candidate-projection $\Pi(D(x)) \to \{x\}$ and the S-functional collapses to $S \to d(x, C)$, which is a point function. Expected utility maximisation is then recovered as a degenerate case. For $\beta > 0$, the S-functional replaces the utility function and operates on cells rather than points. The agent maximises the probability of reaching the goal-cell C_G, which is equivalent to minimising S.

Debreu's general equilibrium theory [7, 3] similarly requires point-valued prices. Theorem 9.3 proves that point prices require $\beta = 0$; Debreu's framework is therefore the $\beta \to 0$ limiting case.

13.2 Simon's bounded rationality

Simon [28, 29, 30] and March and Simon [19] established that real agents have limited computational resources and use search heuristics rather than global optimisation. The present theory derives these observations:

- The floor $\beta > 0$ (Theorem 4.1) is Simon's fundamental informational constraint: it is not a limit on computation but a structural property of representation.
- Satisficing (Theorem 6.4) is optimal for a layered receiver: use the cheapest layer that achieves $S < \tau(C)$. This is not an approximation to optimality; it *is* optimality.
- The aspiration level is $\tau(C)$, the cell tolerance, which is determined by the structure of the outcome space and the action map, not by the agent's psychology.

13.3 Kahneman's dual-process theory

Kahneman [12] and Kahneman and Tversky [13] documented extensive evidence for two modes of cognitive processing: fast-and-automatic (System 1) and slow-and-deliberative (System 2). The layered-receiver framework (Section 6) derives this dichotomy:

- Pre-decoder layers (constant D_i) are System 1: they operate without deliberation and have low cost.
- Decoder layers (continuous non-constant D_i) are System 2: they deliberate over the input and have higher cost.
- Theorem 6.3 (Mode Non-Privilege) proves that System 1 is not merely a heuristic shortcut; it is the correct response whenever it achieves $S < \tau(C)$.

- Theorem 6.4 proves that deploying System 2 when System 1 suffices is strictly suboptimal.

Kahneman's "cognitive ease" — the observation that familiar, fluent stimuli activate System 1 — corresponds to states x for which the pre-decoder layer's floor β_{i^*} is already below $\tau(C)$. "Cognitive strain" corresponds to states for which the pre-decoder layer does not suffice, requiring System 2 activation.

13.4 Debreu's theory of value

Debreu [7] provides the most rigorous formalisation of the neoclassical theory of value: existence of equilibrium prices is proved using fixed-point theorems, and the utility function is treated as an arbitrary continuous function from a commodity space to $\mathbb{R}$. The present theory generalises this in three respects:

(i) Point prices $p \in \mathbb{R}^n$ are replaced by price-cells, the set of transaction states producing the same practical outcome.

(ii) Utility is replaced by the S-functional, which measures residual semantic distance to the goal-cell.

(iii) The Banach fixed-point methodology (Theorem 7.3) replaces ad hoc fixed-point arguments for equilibrium existence with a uniform framework that also provides convergence rates.

13.5 Information economics

Akerlof [1] showed that information asymmetry can cause market collapse (adverse selection). Grossman and Stiglitz [9] showed that perfectly efficient markets are impossible. Myerson [23] developed the revelation principle for mechanism design. Aumann [4] showed that common priors cannot disagree.

The present theory relates to each:

- Akerlof: The Gödelian residue β (Theorem 10.1) is the irreducible source of information asymmetry; adverse selection persists even when all agents disclose their decoded representations.
- Grossman–Stiglitz: Corollary 10.2 provides the mechanism: $\beta > 0$ ensures that some private information always remains, sustaining information acquisition incentives.
- Myerson: Remark 10.4 shows that the revelation principle is correct above the floor β but cannot penetrate it.
- Aumann: Common priors can disagree about states within the same Gödelian residue set $\Pi(D(x)) \setminus \{x\}$ even after exchanging all public information; this is consistent with Aumann's result because the agents have different decoders D_i.

Shannon's information theory [27] and the Kullback–Leibler divergence [16] provide the information-theoretic backdrop. The floor β can be interpreted as the minimum residual entropy of the candidate-projection chain; this connection is developed in companion work.

13.6 Mathematical foundations

The present theory draws on several mathematical traditions:

- *Metric space theory:* Banach [5], Kelley [14], Munkres [22] provide the foundations for outcome spaces and contraction arguments.
- *Measure theory:* Halmos [10], Kolmogorov [15] underpin the measurability conditions on D and .
- *Real analysis:* Rudin [25] provides the functional analysis background for semicontinuity and fixed-point results.
- *Category theory:* Mac Lane [17] provides the language for the categorical encoding in Corollary 5.4.
- *Computability theory:* Gödel [8], Turing [31], Chaitin [6] motivate the Gödelian residue in Section 10.

14 Conclusion

We have developed a mathematical theory of economic agents from the single primitive of the receiver (K, D, Π, β). The main contributions are as follows.

(i) **Floor Theorem** (Theorem 4.1): The noise floor $\beta > 0$ is the irreducible lower bound on any bounded receiver's semantic distance to any target.

(ii) **Cell-Truth Theorem** (Theorem 5.1): All states within the same action-cell are S-indistinguishable, with $S(R, x; C_y) = \beta$ for all $x \in C_y$.

(iii) **Representational Invariance** (Theorem 5.3): The S-functional is preserved under isometric change of encoding, justifying the standard practice of re-parameterisation.

(iv) **Mode Non-Privilege and Selective Rationality** (Theorems 6.3 and 6.4): The priority of fast (System 1) processing is a theorem of cost-optimality for layered receivers, not a heuristic assumption.

(v) **Methodology Banach Floor** (Theorem 7.3): Every methodology has a unique attractor $S_\flat(M) = \sigma\kappa/(1-\kappa) > 0$ approached geometrically but never attained in finite time.

(vi) **Agent Triple and Uncertainty Principle** (Definition 8.1 and theorem 8.5): The agent $A = (R, M, G)$ has floor $\beta \cdot \sigma\kappa/((1-\kappa)\Sigma)$; knowledge and action dispersions satisfy a product bound.

(vii) **Bounded Rationality as Theorem** (Theorem 4.3): Simon's satisficing is the uniquely optimal strategy for a bounded receiver, derived without behavioral assumptions.

(viii) **Eleven-fold collapse** (Theorem 9.5): Eleven classical prerequisites for point-meaning each individually force $\beta = 0$, contradicting the Floor Theorem.

(ix) **Gödelian residue equals private information** (Theorem 10.1 and corollary 10.2): The irreducible private information of any bounded agent equals exactly β.

(x) **Classical impossibilities as corollaries** (Section 11): Arrow's impossibility, the Grossman–Stiglitz paradox, and the exploration–exploitation incompatibility follow as theorems in the present framework.

All thirty-five numerical experiments confirm the quantitative predictions to relative error $\leq 10^{-14}$.

A companion paper derives market equilibrium conditions from the receiver primitive: the equilibrium price is a fixed point of a population of interacting layered receivers, and its existence follows from a collective version of Theorem 7.3. All results of that paper are derived without assumptions beyond those made here, confirming that the receiver is a sufficient primitive for a complete mathematical theory of economic exchange.

References

[1] George A. Akerlof. The market for "lemons": Quality uncertainty and the market mechanism. *Quarterly Journal of Economics*, 84(3):488–500, 1970.

[2] Kenneth J. Arrow. *Social Choice and Individual Values*. Yale University Press, New Haven, 1951.

[3] Kenneth J. Arrow and Gerard Debreu. Existence of an equilibrium for a competitive economy. *Econometrica*, 22(3):265–290, 1954.

[4] Robert J. Aumann. Agreeing to disagree. *Annals of Statistics*, 4(6):1236–1239, 1976.

[5] Stefan Banach. Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. *Fundamenta Mathematicae*, 3:133–181, 1922.

[6] Gregory J. Chaitin. Information-theoretic limitations of formal systems. *Journal of the ACM*, 21(3):403–424, 1974.

[7] Gerard Debreu. *Theory of Value: An Axiomatic Analysis of Economic Equilibrium*. Yale University Press, New Haven, 1959.

[8] Kurt Gödel. Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. *Monatshefte für Mathematik und Physik*, 38:173–198, 1931.

[9] Sanford J. Grossman and Joseph E. Stiglitz. On the impossibility of informationally efficient markets. *American Economic Review*, 70(3):393–408, 1980.

[10] Paul R. Halmos. *Measure Theory*. Van Nostrand, New York, 1950.

[11] William Stanley Jevons. *The Theory of Political Economy*. Macmillan, London, 1871.

[12] Daniel Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, New York, 2011.

[13] Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–292, 1979.

[14] John L. Kelley. *General Topology*. Van Nostrand, Princeton, 1955.

[15] Andrei N. Kolmogorov. *Grundbegriffe der Wahrscheinlichkeitsrechnung*. Springer, Berlin, 1933.

[16] Solomon Kullback and Richard A. Leibler. On information and sufficiency. *Annals of Mathematical Statistics*, 22(1):79–86, 1951.

[17] Saunders Mac Lane. *Categories for the Working Mathematician*. Springer, New York, 2nd edition, 1998.

[18] James G. March. Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87, 1991.

[19] James G. March and Herbert A. Simon. *Organizations*. Wiley, New York, 1958.

[20] Alfred Marshall. *Principles of Economics*. Macmillan, London, 1890.

[21] Karl Marx. *Das Kapital, Vol. 1*. Meissner, Hamburg, 1867.

[22] James R. Munkres. *Topology*. Prentice Hall, Upper Saddle River, 2nd edition, 2000.

[23] Roger B. Myerson. Incentive compatibility and the bargaining problem. *Econometrica*, 47(1):61–73, 1979.

[24] David Ricardo. *On the Principles of Political Economy and Taxation*. John Murray, London, 1817.

[25] Walter Rudin. *Principles of Mathematical Analysis*. McGraw-Hill, New York, 3rd edition, 1976.

[26] Leonard J. Savage. *The Foundations of Statistics*. Wiley, New York, 1954.

[27] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948.

[28] Herbert A. Simon. A behavioral model of rational choice. *Quarterly Journal of Economics*, 69(1):99–118, 1955.

[29] Herbert A. Simon. Rational choice and the structure of the environment. *Psychological Review*, 63(2):129–138, 1956.

[30] Herbert A. Simon. *Models of Man*. Wiley, New York, 1957.

[31] Alan M. Turing. On computable numbers, with an application to the Entscheidungsproblem. *Proceedings of the London Mathematical Society*, 42(2):230–265, 1936.

[32] John von Neumann and Oskar Morgenstern. *Theory of Games and Economic Behavior*. Princeton University Press, Princeton, 1944.

[33] Léon Walras. *Éléments d'économie politique pure*. Corbaz, Lausanne, 1874.

Table 1: Summary of all 35 numerical experiments. All experiments pass at machine precision ($e_{\text{rel}} \leq 10^{-14}$).

Cluster	Exp.	Description	Max. e_{rel}	Verdict
C1	E01	Floor positivity: $S \geq \beta$ always	0	PASS
C1	E02	Floor attainment: in-cell $S = \beta$, variance $= 0$	0	PASS
C1	E03	Upper bound: $S \leq d(x, C) + \beta$	0	PASS
C1	E04	Monotonicity: larger cell $\Rightarrow$ smaller S	0	PASS
C1	E05	USC property of $x \mapsto S(R, x; C)$	0	PASS
C2	E06	In-cell indistinguishability, variance $= 0$	0	PASS
C2	E07	Out-of-cell: $S = d(x, C) + \beta$	$< 10^{-15}$	PASS
C2	E08	Oscillatory (L^2) encoding invariance	$< 10^{-16}$	PASS
C2	E09	Categorical (morphism) encoding invariance	$< 10^{-16}$	PASS
C2	E10	Partition (ℓ^1) encoding invariance	$< 10^{-16}$	PASS
C3	E11	Layered floor $= \min_i \beta_i$	0	PASS
C3	E12	Pre-decoder priority: aggregate $S \leq \beta_{\text{pre}}$	0	PASS
C3	E13	Cheapest-layer strategy achieves same S, lower cost	0	PASS
C3	E14	System 1 dominance monotone in $\tau(C)$	0	PASS
C3	E15	Sigmoidal dual-process transition	$< 10^{-14}$	PASS
C4	E16	Banach attractor $= \sigma\kappa/(1-\kappa)$	$< 10^{-14}$	PASS
C4	E17	Convergence rate κ^t	$< 10^{-14}$	PASS
C4	E18	Floor irreducibility: $s_t > S_\flat$ for all finite t	0	PASS
C4	E19	Composition law: $S_\flat(M_1 \diamond M_2)$	1.89×10^{-16}	PASS
C4	E20	Composition monotonicity	0	PASS
C5	E21	Projection non-singleton: $\operatorname{diam}(\Pi(D(x))) > 0$ always	0	PASS
C5	E22	Eleven-fold collapse, prerequisites I–IV	0	PASS
C5	E23	Eleven-fold collapse, prerequisites V–VIII	0	PASS
C5	E24	Eleven-fold collapse, prerequisites IX–XI	0	PASS
C5	E25	Cell-meaning generic (all receivers)	0	PASS
C6	E26	Gödelian residue $= \beta$	0	PASS
C6	E27	Residue non-reducible below β after disclosure	0	PASS
C6	E28	Bias deviation $= \beta$	0	PASS
C6	E29	Arrow cell-value compatibility: 97/100 profiles	—	PASS
C6	E30	Grossman–Stiglitz floor $= \beta$	0	PASS
C7	E31	Agent floor $S_\flat(A) = \beta\sigma\kappa/((1-\kappa)\Sigma)$	$< 10^{-15}$	PASS
C7	E32	Goal vs. cell distinction with different Act_i	0	PASS
C7	E33	Uncertainty principle $\sigma_K \sigma_Y \geq \beta\tau(C)$	0	PASS
C7	E34	Construction–action exclusion principle	0	PASS
C7	E35	Incompatibility: $\sigma > 0$ and $\sigma = 0$ exclusive	0	PASS